

DEPARTMENT OF MECHANICAL ENGINEERING

MECHATRONICS & IOT 24MEC51

EXERCISE NO: 3

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MECHATRONICS & IoT

ASSIGNMENT REPORT – 3

TITLE:

Design and simulate Smart Elevator Automation and Human Detection System using PIR Sensor. The system automatically detects human presence and controls elevator operations for improved safety and automation.

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Project Overview:

The Smart Elevator Automation and Human Detection System is designed to improve elevator safety and automation by detecting human presence using a PIR (Passive Infrared) sensor.

The system uses a microcontroller such as ESP8266/Arduino to process the PIR sensor signal and control the elevator operation. The PIR sensor detects infrared radiation produced by humans and identifies whether a person is present near or inside the elevator.

Based on the detected human presence, the system can control elevator-related functions such as door operation, elevator activation, floor indication, and safety alerts.

The proposed system provides a simple and low-cost prototype for demonstrating human detection, automatic elevator operation, and safety monitoring.

Problem Statement

Conventional elevator systems may require continuous user interaction and do not always provide additional human-presence-based safety features.

Potential problems include:

- Elevator operation without confirming human presence.
- Doors closing when a person is entering or exiting.
- Unnecessary elevator operation.
- Difficulty detecting human presence in certain situations.
- Lack of additional safety monitoring.
- Manual operation requirements.

Therefore, a smart elevator system is proposed that uses a PIR sensor to detect human presence and automatically control elevator operations.

Proposed Solution

The proposed system consists of:

- PIR sensor
- ESP8266/Arduino microcontroller
- Servo motor for elevator-door simulation
- LEDs
- Buzzer
- Push buttons for floor selection
- Optional display

The PIR sensor detects human movement/presence and sends a digital signal to the microcontroller.

The controller processes this signal and determines whether a person is present.

When human presence is detected:

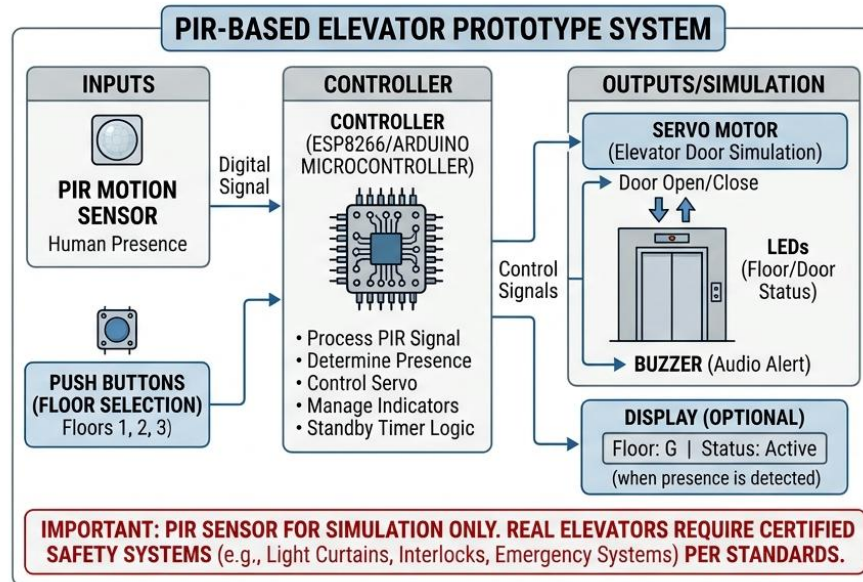
- The elevator system becomes active.
- The door can be opened automatically.
- The selected floor operation can be enabled.
- A presence indicator can be activated.

When no person is detected for a predefined period:

- The elevator can enter standby mode.
- The door can close.
- The system can return to its idle state.

PIR sensors detect changes in infrared radiation/motion and are not reliable as the sole safety sensor for real passenger elevators. A real elevator requires certified safety systems, door-edge/light-curtain sensing, interlocks, overspeed protection, emergency systems, and compliance with applicable elevator standards.

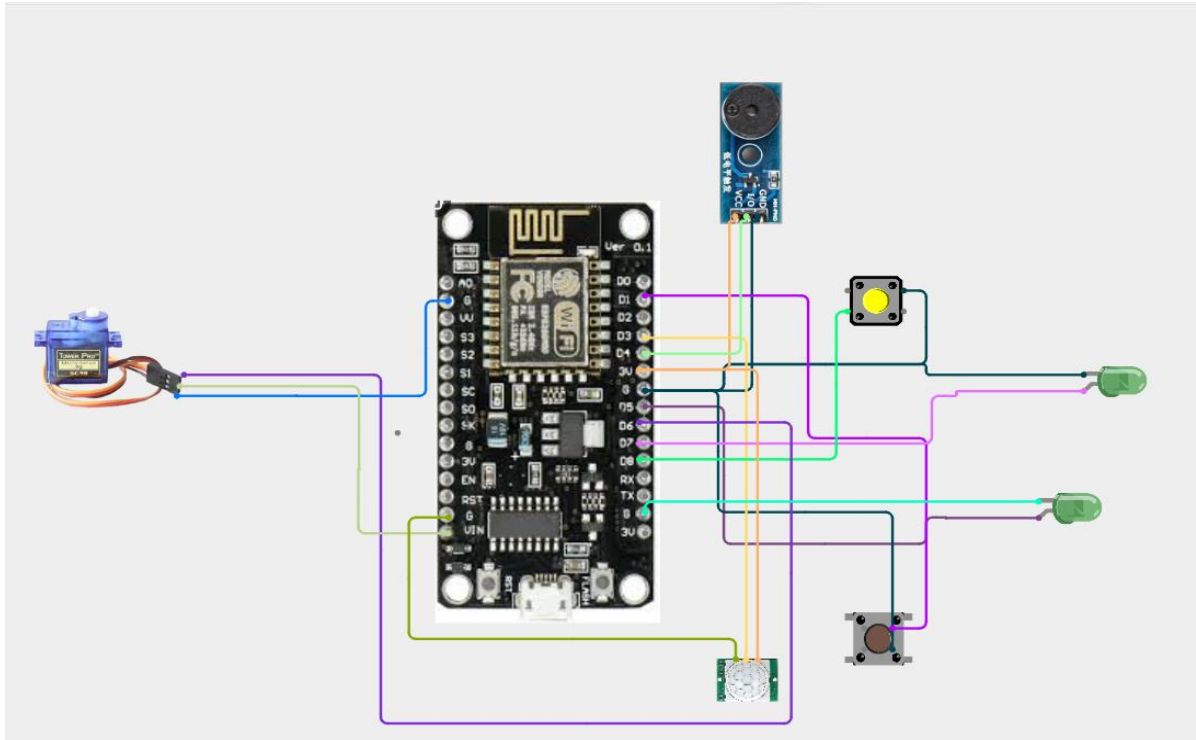
System Architecture



Circuit Table

S.No.	Component	Pin/Terminal	ESP8266 Connection
1	PIR Sensor	VCC	5V/VIN or suitable supply
2	PIR Sensor	GND	GND
3	PIR Sensor	OUT	D5 (GPIO14)
4	Servo Motor	Signal	D4 (GPIO2)
5	Servo Motor	VCC	External 5V supply
6	Servo Motor	GND	Common GND
7	Green LED	Anode	D1 through 220 Ω
8	Green LED	Cathode	GND
9	Red LED	Anode	D2 through 220 Ω
10	Red LED	Cathode	GND
11	Yellow LED	Anode	D6 through 220 Ω
12	Yellow LED	Cathode	GND
13	Buzzer	Positive	D7 through suitable driver
14	Buzzer	Negative	GND
15	Floor Button 1	One terminal	D0
16	Floor Button 2	One terminal	D3
17	Floor Button 3	One terminal	D8

Circuit Design



Algorithm

1. START
2. Initialize ESP8266/Arduino.
3. Initialize PIR sensor.
4. Initialize servo motor.
5. Initialize LEDs and buzzer.
6. Initialize floor-selection buttons.
7. Set elevator to IDLE/STANDBY mode.

8. Read PIR sensor.
9. Is human presence detected?
10. If NO:
 - Keep elevator in standby mode.
 - Keep door closed.
 - Continue monitoring.
11. If YES:
 - Activate elevator system.
 - Turn ON presence indicator.
 - Open elevator door.
12. Check floor-selection buttons.
13. If a floor is selected:
 - Record selected floor.
 - Close elevator door.
 - Simulate elevator movement.
 - Display selected floor.
14. After reaching the selected floor:
 - Stop elevator.
 - Open the door.
15. Monitor PIR sensor near the door.
16. If human presence is detected:
 - Keep the door open.
17. If no human presence is detected for the predefined delay:
 - Close the door.
18. Return to standby mode.
19. Repeat the process continuously.
20. STOP.

Coding

```
#include <ESP8266WiFi.h>
```

```
#include "AdafruitIO_WiFi.h"
```

```
#include <Servo.h>
```

```
//=====
```

```
// WIFI
```

```
//=====
```

```
#define WIFI_SSID "alexa 007"
```

```
#define WIFI_PASS "7871976757"
```

```
//=====
```

```
// ADAFRUIT IO
```

```
//=====
```

```
#define IO_USERNAME "madhan_2006"
```

```
#define IO_KEY      "aio_Ydvh727M22rGsea2zvl7Apeq6Ui1"
```

```
AdafruitIO_WiFi io(IO_USERNAME, IO_KEY, WIFI_SSID, WIFI_PASS);
```

```
// Feeds
```

```
AdafruitIO_Feed *humanFeed = io.feed("human-detected");
```

```
AdafruitIO_Feed *doorFeed = io.feed("door-status");
```

```
AdafruitIO_Feed *floorFeed = io.feed("floor");
```

```
AdafruitIO_Feed *buzzerFeed = io.feed("buzzer-status");
```

```
AdafruitIO_Feed *systemFeed = io.feed("system-status");
```

```
//=====
```

```
// SERVO
```

```
//=====
```

```
Servo elevatorServo;
```

```
//=====
```

```
// PIN CONNECTIONS
```

```
//=====
```

```
#define PIR_PIN D1
```

```
#define SERVO_PIN D2
```

```
#define FLOOR1_BUTTON D5
```

```
#define FLOOR2_BUTTON D6
```

```
#define FLOOR1_LED D7
```

```
#define FLOOR2_LED D0
```

```
#define BUZZER D4
```

```
//=====
```

```
// SERVO POSITIONS
```

```
//=====
```

```
#define DOOR_CLOSED 0
```

```
#define DOOR_OPEN 90
```

```
//=====
```

```
bool systemReady = true;
```

```
//=====
```

```
void setup()
```

```
{  
  
  Serial.begin(115200);  
  
  delay(1000);  
  
  pinMode(PIR_PIN, INPUT);  
  
  pinMode(FLOOR1_BUTTON, INPUT_PULLUP);  
  pinMode(FLOOR2_BUTTON, INPUT_PULLUP);  
  
  pinMode(FLOOR1_LED, OUTPUT);  
  pinMode(FLOOR2_LED, OUTPUT);  
  
  pinMode(BUZZER, OUTPUT);  
  
  digitalWrite(FLOOR1_LED, LOW);  
  digitalWrite(FLOOR2_LED, LOW);  
  digitalWrite(BUZZER, LOW);  
  
  elevatorServo.attach(SERVO_PIN);  
  elevatorServo.write(DOOR_CLOSED);
```

```
Serial.println("Connecting to Adafruit IO...");
```

```
io.connect();
```

```
while(io.status() < AIO_CONNECTED)
```

```
{
```

```
    Serial.print(".");
```

```
    delay(500);
```

```
}
```

```
Serial.println();
```

```
Serial.println("Connected!");
```

```
// Initial Dashboard Values
```

```
humanFeed->save("NO");
```

```
doorFeed->save("CLOSED");
```

```
floorFeed->save("READY");
```

```
buzzerFeed->save("OFF");
```

```
systemFeed->save("READY");
```

```
Serial.println("Smart Elevator Ready");
```

```
}
```

```
//=====
```

```
void loop()
```

```
{
```

```
  io.run();
```

```
  // Human Detection
```

```
  if(systemReady && digitalRead(PIR_PIN)==HIGH)
```

```
  {
```

```
    systemReady=false;
```

```
    Serial.println("HUMAN DETECTED");
```

```
    humanFeed->save("YES");
```

```
    systemFeed->save("WAITING");
```

```
    doorFeed->save("OPEN");
```

```
    elevatorServo.write(DOOR_OPEN);
```

```
delay(1000);
```

```
Serial.println("Select Floor");
```

```
}
```

```
if(!systemReady)
```

```
{
```

```
if(digitalRead(FLOOR1_BUTTON)==LOW)
```

```
{
```

```
    delay(50);
```

```
    if(digitalRead(FLOOR1_BUTTON)==LOW)
```

```
    {
```

```
        floor1Operation();
```

```
    }
```

```
}
```

```
else if(digitalRead(FLOOR2_BUTTON)==LOW)
```

```
{
```

```
    delay(50);
```

```
    if(digitalRead(FLOOR2_BUTTON)==LOW)

    {

        floor2Operation();

    }

}

delay(20);
}

//=====

// FLOOR 1

//=====

void floor1Operation()

{

    Serial.println("FLOOR 1 SELECTED");

    floorFeed->save("FLOOR 1");

    systemFeed->save("MOVING");

    digitalWrite(FLOOR1_LED,HIGH);
```



```
Serial.println("Buzzer ON");
```

```
buzzerFeed->save("ON");
```

```
digitalWrite(BUZZER,HIGH);
```

```
delay(500);
```

```
digitalWrite(BUZZER,LOW);
```

```
buzzerFeed->save("OFF");
```

```
Serial.println("Moving to Floor 1");
```

```
delay(2000);
```

```
Serial.println("Arrived");
```

```
elevatorServo.write(DOOR_CLOSED);
```

```
doorFeed->save("CLOSED");
```

```
delay(1000);
```

```
digitalWrite(FLOOR1_LED,LOW);
```

```
while(digitalRead(FLOOR1_BUTTON)==LOW)
```

```
{
```

```
    io.run();
```

```
    delay(20);
```

```
}
```

```
while(digitalRead(PIR_PIN)==HIGH)
```

```
{
```

```
    io.run();
```

```
    delay(50);
```

```
}
```

```
humanFeed->save("NO");
```

```
floorFeed->save("READY");
```

```
systemFeed->save("READY");
```

```
systemReady=true;
```

```
Serial.println("Ready Again");
```

```
}
```

```
//=====
```

```
// FLOOR 2
```

```
//=====
```

```
void floor2Operation()
```

```
{
```

```
    Serial.println("FLOOR 2 SELECTED");
```

```
    floorFeed->save("FLOOR 2");
```

```
    systemFeed->save("MOVING");
```

```
    digitalWrite(FLOOR2_LED,HIGH);
```

```
    buzzerFeed->save("ON");
```

```
    digitalWrite(BUZZER,HIGH);
```

```
    delay(500);
```

```
    digitalWrite(BUZZER,LOW);
```

```
    buzzerFeed->save("OFF");
```

```
Serial.println("Moving to Floor 2");
```

```
delay(2000);
```

```
Serial.println("Arrived");
```

```
elevatorServo.write(DOOR_CLOSED);
```

```
doorFeed->save("CLOSED");
```

```
delay(1000);
```

```
digitalWrite(FLOOR2_LED,LOW);
```

```
while(digitalRead(FLOOR2_BUTTON)==LOW)
```

```
{
```

```
  io.run();
```

```
  delay(20);
```

```
}
```

```
while(digitalRead(PIR_PIN)==HIGH)
```

```
{
```

```
  io.run();
```

```
  delay(50);
```

```
}
```

```
humanFeed->save("NO");  
floorFeed->save("READY");  
systemFeed->save("READY");  
  
systemReady=true;  
  
Serial.println("Ready Again");  
}
```

System Working

Step 1 – Human Detection

The PIR sensor continuously monitors the area around the elevator.

When a person enters the detection zone, the PIR sensor produces a HIGH signal.

Step 2 – Controller Processing

The ESP8266 receives the signal and identifies human movement.

The controller activates the elevator system.

Step 3 – Door Automation

The servo motor opens the elevator door.

Step 4 – Floor Selection

The passenger selects the required floor using a push button.

The controller records the selected floor.

Step 5 – Elevator Movement Simulation

The system simulates elevator movement using LEDs, display indicators, or a motor mechanism.

Step 6 – Arrival

After the simulated movement is completed:

- Elevator stops.
- Door opens.
- Passenger can exit.

Step 7 – Standby

When no human presence is detected for the predefined time, the door closes and the elevator returns to standby mode.

Bills Of Material

S.N o.	Component	Specification	Quantity	Purpose
1	ESP8266 NodeMCU	Wi-Fi Microcontroller	1	Main Controller
2	PIR Sensor	HC-SR501	1	Human Detection
3	Servo Motor	SG90	1	Door Control
4	Push Buttons	Tactile Switch	3	Floor Selection
5	Green LED	5 mm	1	Active Indication
6	Red LED	5 mm	1	Alert Indication
7	Yellow LED	5 mm	1	Standby/Movement
8	Resistor	220 Ω	3	LED Protection
9	Buzzer	3.3/5 V	1	Audible Alert
10	Breadboard	Standard	1	Circuit Assembly
11	Jumper Wires	Male-Male	1 Set	Connections
12	USB Cable	Micro USB	1	Programming/Power

Prototype Implementation

The prototype is assembled using an ESP8266 NodeMCU, PIR sensor, servo motor, push buttons, LEDs, buzzer, breadboard, and jumper wires.

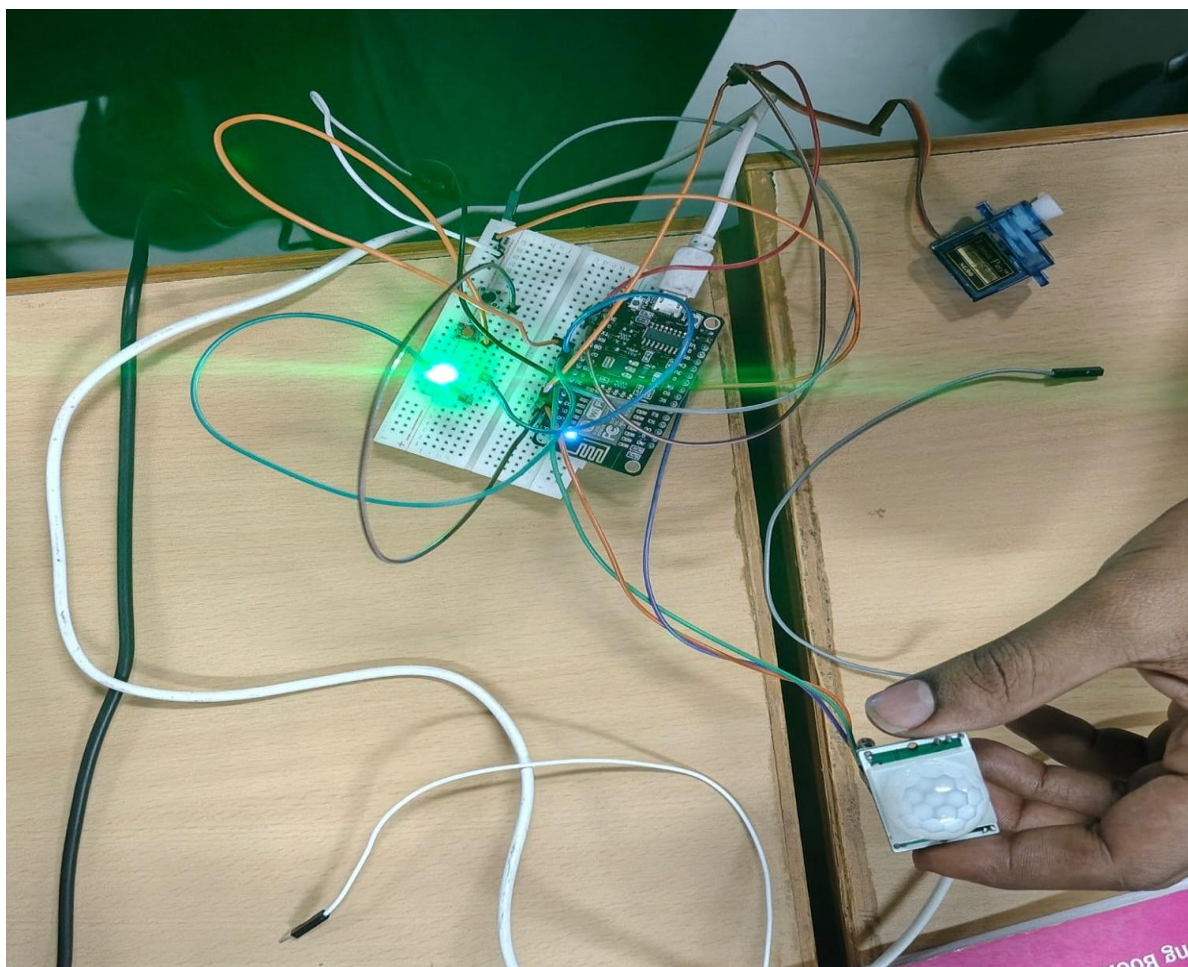
The PIR sensor is positioned near the elevator entrance to detect human movement.

When a person is detected, the ESP8266 activates the elevator and commands the servo motor to open the door.

The passenger selects a floor using one of the push buttons. The controller then closes the door and simulates elevator movement.

After the simulated travel period, the elevator stops at the selected floor and the door opens.

The system then waits for the passenger to exit before returning to standby mode.



Results & Performance Analysis

The prototype successfully demonstrates human detection and basic elevator automation.

Parameter	Result
Human Detection	Yes
PIR Sensor Monitoring	Yes
Automatic Door Control	Yes
Floor Selection	Yes
Elevator Movement Simulation	Yes
Status Indication	Yes
Audible Alert	Yes
Automatic Standby	Yes

The PIR sensor provides a simple method of detecting human movement. The microcontroller processes the sensor signal and controls the elevator-door mechanism.

The servo motor provides an effective way to simulate elevator door movement in a small-scale prototype.

```
Human Detected
Door OPEN
Floor 2 Selected
Moving to Floor 2...
Arrived at Floor 2
Door CLOSED
Floor 2 Complete
Human Detected
Door OPEN
Floor 2 Selected
Moving to Floor 2...
Arrived at Floor 2
Door CLOSED
Floor 2 Complete
Human Detected
Door OPEN
Floor 2 Selected
Moving to Floor 2...
Arrived at Floor 2
Door CLOSED
Floor 2 Complete
Human Detected
Door OPEN
Floor 2 Selected
Moving to Floor 2...
```

```
Door OPEN
Human Detected
Door OPEN
Human Detected
Door OPEN
Human Detected
Door OPEN
Floor 2 Selected
Moving to Floor 2...
Arrived at Floor 2
Door CLOSED
Floor 2 Complete
Human Detected
Door OPEN
Floor 2 Selected
Moving to Floor 2...
Arrived at Floor 2
Door CLOSED
Floor 2 Complete
Human Detected
Door OPEN
Floor 2 Selected
Moving to Floor 2...
```

OFF

CLOSED

FLOOR 1

MOVING

YES